

10/2/2021

Ex1 $V_{C2} = 11V$

$$I_{C2} = \frac{V_{C2}}{R_C} = 0.917 \text{ mA}$$

$$V_X = V_{BE1} + I_{B1} R_F = 15.84 \text{ V}$$

$$I_{C1} = I_{E1} + \frac{I_{C2}}{\beta_F} = 1.009 \text{ mA}$$

$$V_{CE1} = V_X + V_{CE2} = 15.35 \text{ V}$$

$$V_{CE2} = V_{C2} - V_X = -4.83 \text{ V}$$

sbagliato questo
con lo sbagliato
tutti i risultati

$$R_E = \frac{V_{C2} - V_X}{I_{B1} + \left(1 + \frac{1}{\beta_F}\right) I_{C2}} = 8.72 \text{ k}\Omega$$



small signal analysis

C_S, C_C shorted, C_E open

$$A_v = \frac{V_o}{V_s}$$

amplifier topology:

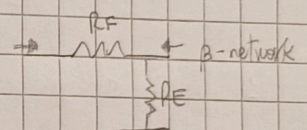
current mixing

(shunt)

Norton equivalent

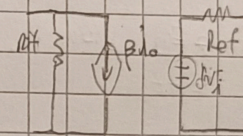
current sampling
series sampling

i_o = current amplifier



$$I_S = \frac{V_S}{R_S}$$

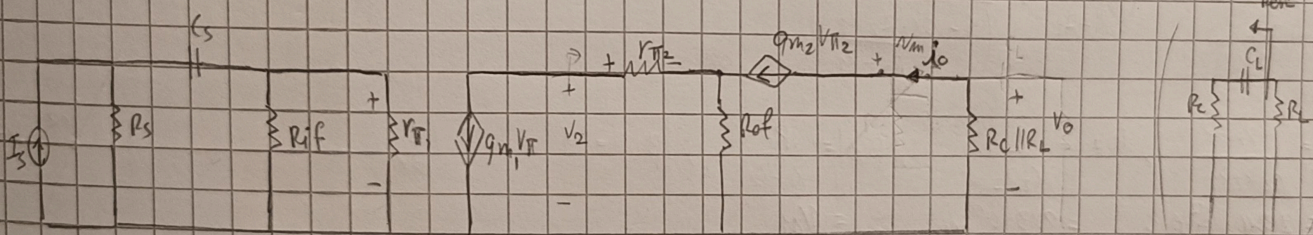
2-port model



$$R_{if} = R_F + R_E = 1509 \text{ k}\Omega$$

$$R_{of} = R_F \parallel R_E = 8.77 \text{ k}\Omega$$

$$\beta = \frac{i_o}{i_i} \bigg|_{V_i=0} = -\frac{R_E}{R_F + R_E} = -5.78 \cdot 10^{-3}$$



$$A_{v_o} = \frac{V_o}{V_s} \bigg|_{\beta=0} = \alpha_i \alpha_{CE} \alpha_{LER}$$

$$g_{m1} = \frac{I_{C1}}{V_T} = 0.04$$

$$g_{m2} = \frac{I_{C2}}{V_T} = 0.0367$$

$$r_{\pi 1} = \frac{\beta_0}{g_{m1}} = 2478 \Omega$$

$$r_{\pi 2} = \frac{\beta_0}{g_{m2}} = 2725 \Omega$$

$$\alpha_{LER} = \frac{i_o}{v_2} = \frac{g_{m2} r_{\pi 2}}{R_{inCE}} = \frac{\beta_0}{R_{inCE}} = 1.126 \cdot 10^{-4} \text{ [V/V]}$$

$$R_{inCE} = r_{\pi 2} + (1 + \beta_0) R_{of} = 887.73 \text{ k}\Omega$$

$$\alpha_{CE} = \frac{v_2}{V_{\pi 1}} = -g_{m1} R_{inCE} = -35509$$

$$\alpha_i = \frac{v_i}{i_s} = R_s \parallel R_i \parallel R_{T1} = 2350 \quad [S]$$

$$A_a = \alpha_i \alpha_{CE} \alpha_{CER} = -9.68 \quad [A/A]$$

$$A_F = \frac{A_a}{1 + \beta A_a} = -168.9 \quad [A/A]$$

$$A_v = \frac{v_o}{v_s} = A_F \cdot \frac{i_s}{v_s} \cdot \frac{v_o}{i_o} = -A_F \cdot \frac{1}{R_s} (R_c \parallel R_L) = +19.7 \quad [V/V]$$

$$R_{out}^a = \frac{N_{in}}{I_{in}} = \infty$$

$$R_{out}^F = R_{out}^a (1 + \beta A_a) = \infty$$

$$R_{out} = (R_{out}^F \parallel R_c \parallel R_L) \parallel R_c = \infty \parallel R_c = R_c = 12 \text{ k}\Omega$$

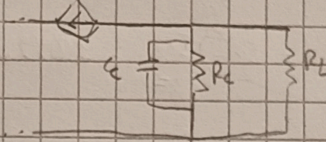
$$C_c = 5 \text{ pF not negligible}$$

OCTC method

$$F_H = \frac{1}{2\pi C_c R_{cc}^{\infty}}$$

$$R_{cc}^{\infty} = R_c \parallel R_L \rightarrow \text{feedback doesn't modify at all the resistance seen from } C_c (v_o = 0)$$

$$F_H = \frac{1}{2\pi C_c R_{cc}^{\infty}} = 5.836 \text{ MHz}$$



Ex2

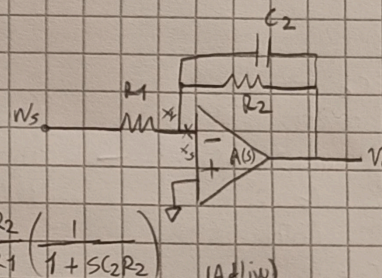
$$R_1 = 10 \text{ k}\Omega, R_2 = 100 \text{ k}\Omega, C_2 = 4.7 \text{ pF}$$

$$\omega_0 = 10^3 \text{ rad/s}$$

$$\omega_1 = 10^4 \text{ rad/s}$$

$$A(s) = \frac{10^5}{(1 + \frac{s}{10^3})(1 + \frac{s}{10^4})}$$

$\frac{v_o}{v_i}$



$$1) \text{ inverting configuration } A = -\frac{Z_2}{Z_1} = -\frac{(R_2 \parallel \frac{1}{sC_2})}{R_1} = -\frac{R_2}{R_1} \left(\frac{1}{1 + sC_2R_2} \right)$$

$$2) T = -\frac{X_f}{X_s} = -A_d(s) \cdot \left(\frac{Z_1}{Z_1 + Z_2} \right) = -A_d(s) \cdot \left(\frac{-R_1}{R_1 + \frac{R_2}{1 + sC_2R_2}} \right)$$

$$\beta = \frac{R_1}{R_1 + R_2} \cdot \frac{(1 + sC_2R_2)}{(1 + sC_2R_2R_1/R_2 + R_1)}$$

$$\omega_p = \frac{R_2 + R_1}{C_2 R_2 R_1}, \omega_z = \frac{1}{C_2 R_2}, |\beta_0| = \frac{R_1}{R_1 + R_2}, |\beta_{\infty}| = 1$$

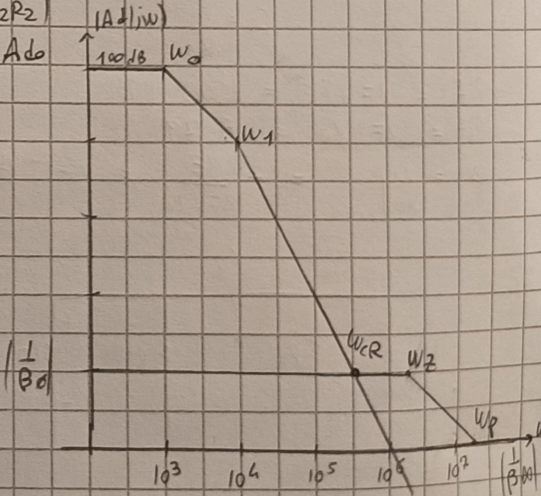
$$= 2.34 \cdot 10^7 \text{ rad/s}$$

$$= 2.13 \cdot 10^6 \text{ rad/s}$$

$$= 1/11$$

$$|\beta_0| = 1/11 = 21 \text{ dB}$$

$$|\beta_{\infty}| = 0 \text{ dB}$$



$$\omega_{CR} : \left\{ \begin{array}{l} A_d(j\omega_{CR}) = A_{u1} \\ |A_d(j\omega_{CR})| = \frac{1}{|\beta_0|} \omega_{CR}^2 \end{array} \right\} \rightarrow A_u = \frac{A_d(j\omega_{CR})}{\omega_{CR}} \rightarrow \omega_{CR} = \sqrt{|\beta_0| A_d(j\omega_{CR}) \omega_{CR}} = 3 \cdot 10^5 \text{ rad/s}$$

$$PM = 180 - \arg\left(\frac{\omega_{CR}}{\omega_0}\right) - \arg\left(\frac{\omega_{CR}}{\omega_1}\right) - \arg\left(\frac{\omega_{CR}}{\omega_p}\right) + \arg\left(\frac{\omega_{CR}}{\omega_z}\right) = 9.38^\circ \text{ low phase margin}$$

3) C_2 such that $PM = 45^\circ$
 we can get a $\pm \Delta\phi = 45^\circ$ by placing the W_2 at W_{CR}

so $W_2' = W_{CR} = \frac{1}{C_2' R_2} \rightarrow C_2' = \frac{1}{W_{CR} R_2} = 33.3 \text{ pF}$

$W_{p1}' = 3.3 \cdot 10^6 \text{ rad/s} \rightarrow PM = 42^\circ$

4) complete block diagram (feedback theory)

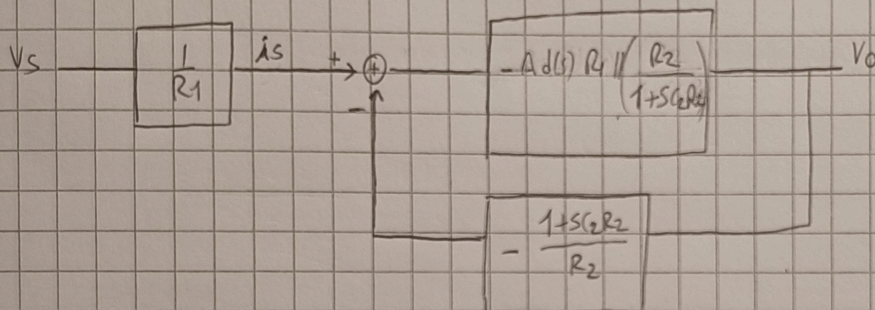
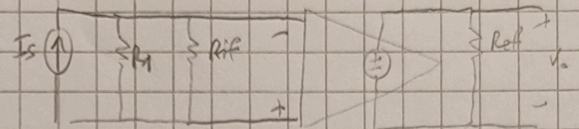
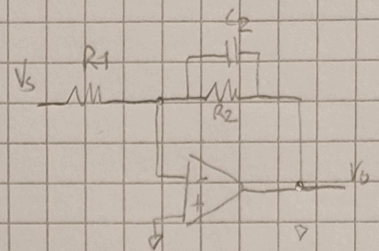
(Notation)

current mixing - voltage sampling

$$R_{if} = R_2 \parallel \frac{1}{sC_2} = \frac{R_2}{1 + sC_2 R_2}$$

$$R_{of} = R_2 \parallel \frac{1}{sC_2} = \frac{R_2}{1 + sC_2 R_2}$$

$$\beta = -\frac{1}{Z_2} = \frac{i_a}{V_o} \Big|_{V_i=0} = -\frac{1}{R_2} = -\frac{1 + sC_2 R_2}{R_2}$$



5) $K_{in}(s) = \frac{1}{R_1}$

6) $A_V = ?$

$w=0$ For $w=0 \rightarrow \beta = -\frac{R_1}{R_1 + R_2}$
 $A(s) = 10^5$

$T = \beta A \alpha = 9090 \gg 1 \rightarrow T \gg 1$

$AF = A_V \approx \frac{1}{\beta} = -11$

1 ora e 15 minuti